WORKSHEET

Least squares regression line

1 For each of the following sets of time series data, use CAS to plot the data, then fit a least-squares regression line.

a Deseasonalised quarterly sales for Colin's football uniform supplies company (figures are in 1000s).

	Q1	Q2	Q3	Q4
2011	22.106	27.897	35.684	22.234
2012	26.986	21.634	22.430	16.676
2013	23.684	23.342	15.293	33.351

b Hamburger Sales from Mac's local takeaway in 2013

Month	Jan	Feb	Mar	Apr	May	June	July	Aug	Sept	Oct	Nov	Dec
Sales	165	183	179	158	176	209	198	1	199	215	179	241

c Number of members registered with Glenworth Car Club

Year	2002	2003	2004	2005	2006	2007	2008	2009	2010
Members	206	187	175	184	166	152	164	148	140

d Average class mark (%) on a weekly revision test

Test	1	2	3	4	5	6	7	8	9	10	11	12
Mark (%)	43	51	57	59	64	70	71	67	73	76	81	79

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2 For each of the following data sets, fit a linear regression line by selecting two data points and finding the equation between them.

a Death toll at a busy highway intersection

Year	2006	2007	2008	2009	2010	2011
No. of deaths	19	21	28	36	33	39

b Number of pigs produced by a small-holding farmer

Year	2005	2006	2007	2008	2009	2010	2011	2012	2013
No. of pigs	2	3	8	11	15	13	15	18	22

c Deseasonalised data on the number of swimmers using the local council pool

	Q1	Q2	Q3	Q4
2009	1645.3	1646.8	1661.6	1709.9
2010	1677.5	1852.6	1726.8	1657.1
2011	1802.0	1622.5	1734.8	1755.2

d Average monthly rainfall in Melbourne

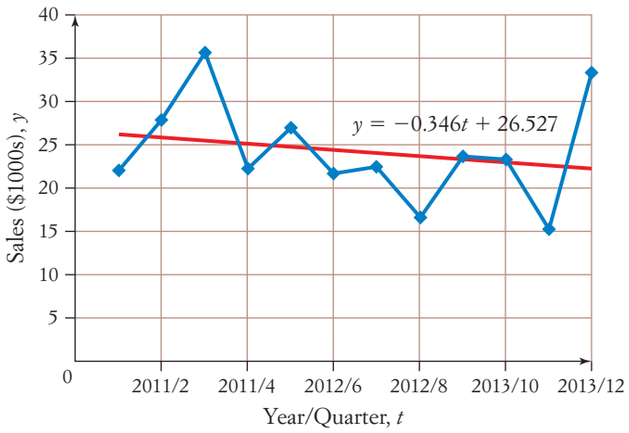
Month	Jan	Feb	Mar	Apr	May	June	July	Aug	Sept	Oct	Nov	Dec
Rainfall (mm)	47.4	48.2	50.5	57.3	56.0	49.3	47.6	50.1	58.0	66.4	60.4	59.4

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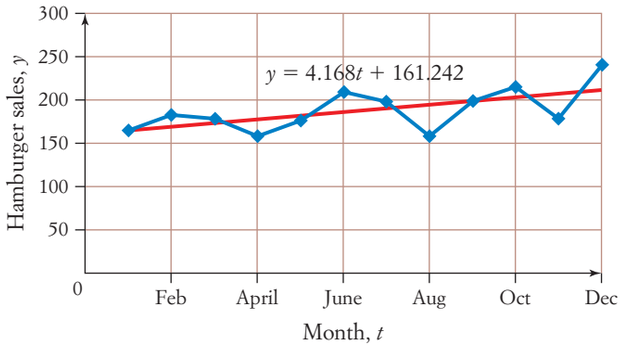
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Answers

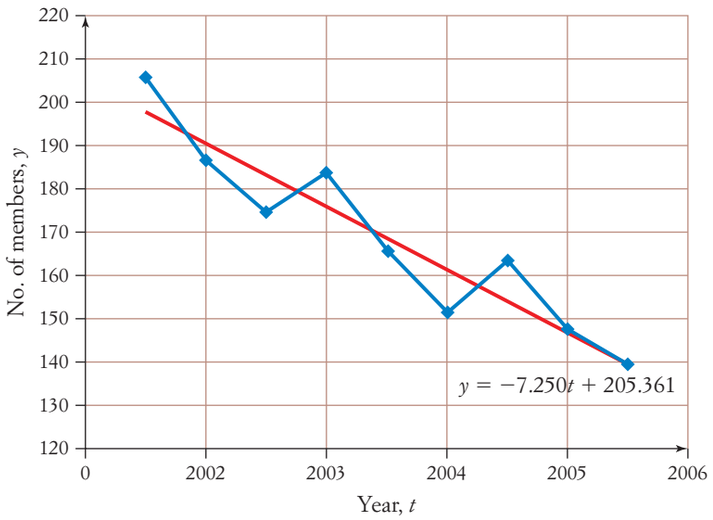
1 a



b

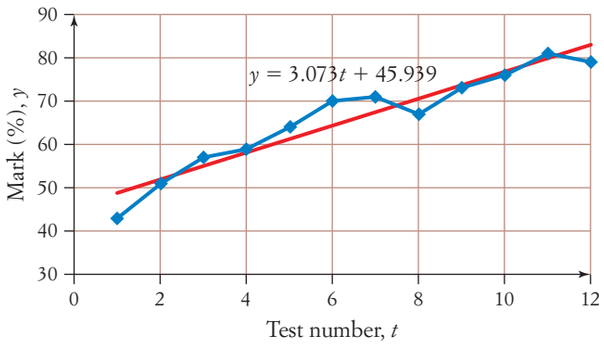


c



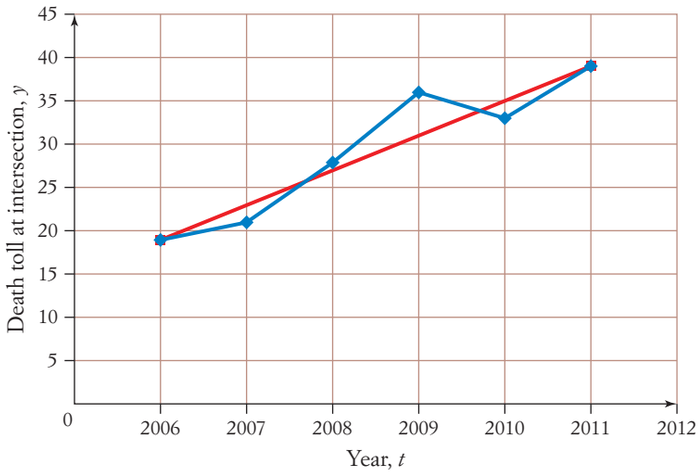
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d

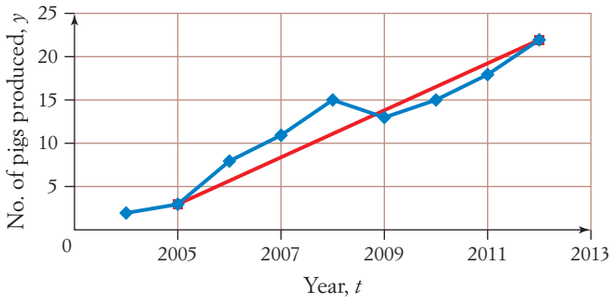


The equation of each regression line will vary, depending on which data values are chosen.

2 a $y = 4t + 15$

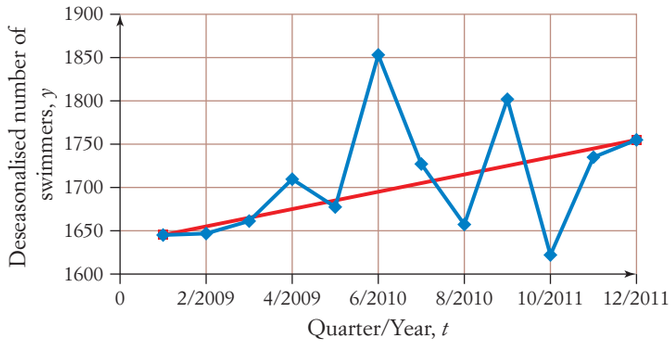


b $y = 2.5t - 0.5$

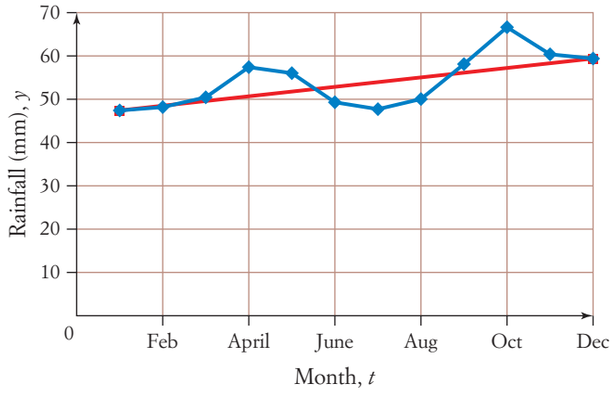


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c $y = 9.991t + 1635.309$



d $y = 1.091t + 46.309$



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